# Pipeline Internals

## Pipeline Internals I

This section documents the artefacts produced by the prose-review run.

```
flowchart TB
    P[projects/templates/template_prose_project/]
    P --> M[manuscript/]
    P --> OUT[output/<br/>tracked public artifacts ·
regenerable]

    M --> M_CFG[config.yaml<br/>single source of
truth]
    M --> M_PRE[preamble.md<br/>LaTeX preamble]
    M --> M_SEC[00_abstract → 04_conclusion
·<br/>05_pipeline_internals · 06_reproducibility]
    M --> M_BIB[references.bib<br/>read-only ·
validated by pipeline]

    OUT -->
O_REP[manuscript_report.json<br/>ManuscriptReport
```

## Pipeline Internals II

```
    OUT --> O_CK[checks.json<br/>list of CheckResult]
    OUT --> O_RR[review_report.md<br/>final markdown
report]
    OUT --> O_DATA[data/manuscript_variables.json]
    OUT -->
O_FIG[figures/<br/>section_word_counts.png
·<br/>readability_metrics.png
·<br/>citation_density.png]
    OUT --> O_RS[run_summary.json]

    classDef d fill:#0f172a,stroke:#0f172a,color:#fff
    classDef src
fill:#1e3a8a,stroke:#0f172a,color:#fff
    classDef gen
fill:#0f766e,stroke:#0f172a,color:#fff
    class P,M,OUT d
    class M_CFG,M_PRE,M_SEC,M_BIB src
    class O_REP,O_CK,O_RR,O_DATA,O_FIG,O_RS gen
```

## Data structures I

```
classDiagram
    class ManuscriptReport {
        +list~FileReport~ files
        +int total_words
        +int total_sentences
        +int total_paragraphs
        +float avg_flesch_reading_ease
        +float avg_flesch_kincaid_grade
        +float avg_gunning_fog
        +list~str~ citation_keys
    }

    class FileReport {
        +str name
        +ProseMetrics metrics
        +StructureReport structure
        +QualityReport quality
```

```
}

class ProseMetrics {
    +int word_count
    +int sentence_count
    +int paragraph_count
    +float flesch_reading_ease
    +float flesch_kincaid_grade
    +float gunning_fog
}

class StructureReport {
    +list~Heading~ headings
    +list~Section~ sections
    +int max_depth
    +bool has_h1
    +bool has_skipped_level
}
```

## Data structures III

```
class QualityReport {
    +int passive_count
    +int hedge_count
    +int citation_count
    +int long_sentence_count
    +float citation_density_per_1000
}

class CheckResult {
    +str name
    +bool passed
    +str message
}

class ProseRunArtifacts {
    +ManuscriptReport manuscript_report
    +list~CheckResult~ checks
```

## Data structures IV

```
    +bool all_passed
}

ManuscriptReport --> "*" FileReport
FileReport --> ProseMetrics
FileReport --> StructureReport
FileReport --> QualityReport
ProseRunArtifacts --> ManuscriptReport
ProseRunArtifacts --> "*" CheckResult
```